

```

=====
57.3/57.3 kB 2.8 MB/s eta 0:00:00
5.4/5.4 MB 45.7 MB/s eta 0:00:00
937.5/937.5 kB 29.7 MB/s eta 0:00:00
25.5/25.5 MB 51.7 MB/s eta 0:00:00
51.5/51.5 kB 5.3 MB/s eta 0:00:00
2.4/2.4 MB 96.1 MB/s eta 0:00:00
8.8/8.8 MB 98.6 MB/s eta 0:00:00
167.2/167.2 kB 17.3 MB/s eta 0:00:00
=====
Using Device: cuda
=====
Mounted at /content/drive

Dataset found at:
/content/drive/MyDrive/Bone_Fracture_Binary_Classification

Seeds:
[42, 52, 62, 72, 82]

Dataset paths:
Train: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/train
Validation: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/val
Test: /content/drive/MyDrive/Bone_Fracture_Binary_Classification/test

Class mapping:
{'fractured': 0, 'not fractured': 1}

Dataset sizes:
Train      : 9246
Validation : 829
Test       : 506

Data loaders created successfully.

Quantum configuration:
Qubits      : 8
Quantum layers : 2
Weight shape : (2, 8, 3)
Downloading: "https://download.pytorch.org/models/efficientnet\_b0\_rwightman-7f5810bc.pth" to /root/.cache/torch/hub/checkpoints/efficientnet_b0_100%|██████████| 20.5M/20.5M [00:00<00:00, 164MB/s]

=====
MODEL PARAMETER INFORMATION
=====
Total parameters      : 4,017,853
Trainable parameters  : 4,017,853
Compression parameters: 10,248
Quantum parameters    : 48
Output parameters     : 9

=====
STARTING FIVE-SEED EXPERIMENT — SEED 42
=====

Random Seed: 42

=====
STARTING DVQC TRAINING FOR SEED 42
=====
Seed 42 | Epoch 1/10: 100%|██████████| 145/145 [17:14<00:00, 7.14s/it, loss=0.618]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 1/10 | Train Loss: 0.6498 | Val Loss: 0.6195 | Train Acc: 87.15% | Val Acc: 97.47%
Seed 42 | Epoch 2/10: 100%|██████████| 145/145 [15:31<00:00, 6.42s/it, loss=0.605]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 2/10 | Train Loss: 0.6146 | Val Loss: 0.6017 | Train Acc: 99.23% | Val Acc: 98.31%
Seed 42 | Epoch 3/10: 100%|██████████| 145/145 [15:27<00:00, 6.40s/it, loss=0.604]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 3/10 | Train Loss: 0.6001 | Val Loss: 0.5878 | Train Acc: 99.56% | Val Acc: 98.43%
Seed 42 | Epoch 4/10: 100%|██████████| 145/145 [15:33<00:00, 6.44s/it, loss=0.571]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 4/10 | Train Loss: 0.5837 | Val Loss: 0.5692 | Train Acc: 99.70% | Val Acc: 98.91%
Seed 42 | Epoch 5/10: 100%|██████████| 145/145 [15:23<00:00, 6.37s/it, loss=0.555]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 5/10 | Train Loss: 0.5658 | Val Loss: 0.5511 | Train Acc: 99.72% | Val Acc: 99.52%
Seed 42 | Epoch 6/10: 100%|██████████| 145/145 [15:33<00:00, 6.44s/it, loss=0.533]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted to RGBA images
warnings.warn(
Seed 42 | Epoch 6/10 | Train Loss: 0.5484 | Val Loss: 0.5381 | Train Acc: 99.77% | Val Acc: 98.07%
Seed 42 | Epoch 7/10: 100%|██████████| 145/145 [15:31<00:00, 6.42s/it, loss=0.535]

```

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 7/10 | Train Loss: 0.5312 | Val Loss: 0.5164 | Train Acc: 99.76% | Val Acc: 99.64%
Seed 42 | Epoch 8/10: 100%|██████████| 145/145 [14:58<00:00, 6.20s/it, loss=0.513]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 8/10 | Train Loss: 0.5129 | Val Loss: 0.4991 | Train Acc: 99.89% | Val Acc: 99.52%
Seed 42 | Epoch 9/10: 100%|██████████| 145/145 [14:54<00:00, 6.17s/it, loss=0.475]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 9/10 | Train Loss: 0.4944 | Val Loss: 0.4797 | Train Acc: 99.94% | Val Acc: 99.88%
Seed 42 | Epoch 10/10: 100%|██████████| 145/145 [15:15<00:00, 6.31s/it, loss=0.465]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 42 | Epoch 10/10 | Train Loss: 0.4761 | Val Loss: 0.4616 | Train Acc: 99.95% | Val Acc: 99.88%
```

Seed 42 completed in 159.94 minutes.

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

RESULTS FOR SEED 42

```
-----
Test Accuracy   : 98.81%
Test Sensitivity: 99.63%
Test Specificity: 97.90%
Test Precision  : 98.16%
Test F1-Score   : 98.89%
Test Error Rate : 1.19%
Test ROC-AUC    : 0.9987
```

STARTING FIVE-SEED EXPERIMENT – SEED 52

Random Seed: 52

STARTING DVQC TRAINING FOR SEED 52

```
Seed 52 | Epoch 1/10: 100%|██████████| 145/145 [15:09<00:00, 6.27s/it, loss=0.579]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 1/10 | Train Loss: 0.6346 | Val Loss: 0.6107 | Train Acc: 59.41% | Val Acc: 81.30%
Seed 52 | Epoch 2/10: 100%|██████████| 145/145 [14:56<00:00, 6.18s/it, loss=0.543]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 2/10 | Train Loss: 0.5670 | Val Loss: 0.5714 | Train Acc: 94.16% | Val Acc: 97.95%
Seed 52 | Epoch 3/10: 100%|██████████| 145/145 [15:11<00:00, 6.29s/it, loss=0.547]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 3/10 | Train Loss: 0.5402 | Val Loss: 0.5471 | Train Acc: 98.95% | Val Acc: 97.71%
Seed 52 | Epoch 4/10: 100%|██████████| 145/145 [15:34<00:00, 6.44s/it, loss=0.522]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 4/10 | Train Loss: 0.5163 | Val Loss: 0.5206 | Train Acc: 99.38% | Val Acc: 98.79%
Seed 52 | Epoch 5/10: 100%|██████████| 145/145 [15:40<00:00, 6.48s/it, loss=0.483]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 5/10 | Train Loss: 0.4933 | Val Loss: 0.4994 | Train Acc: 99.68% | Val Acc: 98.19%
Seed 52 | Epoch 6/10: 100%|██████████| 145/145 [15:15<00:00, 6.31s/it, loss=0.479]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 6/10 | Train Loss: 0.4723 | Val Loss: 0.4765 | Train Acc: 99.65% | Val Acc: 98.55%
Seed 52 | Epoch 7/10: 100%|██████████| 145/145 [15:14<00:00, 6.31s/it, loss=0.426]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 7/10 | Train Loss: 0.4508 | Val Loss: 0.4506 | Train Acc: 99.72% | Val Acc: 99.03%
Seed 52 | Epoch 8/10: 100%|██████████| 145/145 [15:37<00:00, 6.46s/it, loss=0.408]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 8/10 | Train Loss: 0.4293 | Val Loss: 0.4283 | Train Acc: 99.76% | Val Acc: 99.52%
Seed 52 | Epoch 9/10: 100%|██████████| 145/145 [15:42<00:00, 6.50s/it, loss=0.409]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 9/10 | Train Loss: 0.4084 | Val Loss: 0.4068 | Train Acc: 99.81% | Val Acc: 99.52%
Seed 52 | Epoch 10/10: 100%|██████████| 145/145 [15:21<00:00, 6.36s/it, loss=0.399]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 52 | Epoch 10/10 | Train Loss: 0.3883 | Val Loss: 0.3884 | Train Acc: 99.86% | Val Acc: 99.16%
```

Seed 52 completed in 156.64 minutes.

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
```

```
warnings.warn(
```

```
RESULTS FOR SEED 52
```

```
-----  
Test Accuracy   : 98.22%  
Test Sensitivity: 98.13%  
Test Specificity: 98.32%  
Test Precision  : 98.50%  
Test F1-Score   : 98.32%  
Test Error Rate : 1.78%  
Test ROC-AUC    : 0.9967
```

```
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```

```
STARTING FIVE-SEED EXPERIMENT – SEED 62
```

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```

```
Random Seed: 62
```

```
=====
```

```
STARTING DVQC TRAINING FOR SEED 62
```

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```

```
Seed 62 | Epoch 1/10: 100%|██████████| 145/145 [15:32<00:00, 6.43s/it, loss=0.659]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 1/10 | Train Loss: 0.6664 | Val Loss: 0.6229 | Train Acc: 50.18% | Val Acc: 59.35%  
Seed 62 | Epoch 2/10: 100%|██████████| 145/145 [15:19<00:00, 6.34s/it, loss=0.65]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 2/10 | Train Loss: 0.6348 | Val Loss: 0.6068 | Train Acc: 50.18% | Val Acc: 59.35%  
Seed 62 | Epoch 3/10: 100%|██████████| 145/145 [14:57<00:00, 6.19s/it, loss=0.594]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 3/10 | Train Loss: 0.6182 | Val Loss: 0.5904 | Train Acc: 50.18% | Val Acc: 59.35%  
Seed 62 | Epoch 4/10: 100%|██████████| 145/145 [15:20<00:00, 6.35s/it, loss=0.592]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 4/10 | Train Loss: 0.6047 | Val Loss: 0.5797 | Train Acc: 51.29% | Val Acc: 60.80%  
Seed 62 | Epoch 5/10: 100%|██████████| 145/145 [15:01<00:00, 6.22s/it, loss=0.579]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 5/10 | Train Loss: 0.5932 | Val Loss: 0.5697 | Train Acc: 55.98% | Val Acc: 95.05%  
Seed 62 | Epoch 6/10: 100%|██████████| 145/145 [14:56<00:00, 6.18s/it, loss=0.564]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 6/10 | Train Loss: 0.5825 | Val Loss: 0.5595 | Train Acc: 95.09% | Val Acc: 96.98%  
Seed 62 | Epoch 7/10: 100%|██████████| 145/145 [14:57<00:00, 6.19s/it, loss=0.556]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 7/10 | Train Loss: 0.5719 | Val Loss: 0.5492 | Train Acc: 97.80% | Val Acc: 99.52%  
Seed 62 | Epoch 8/10: 100%|██████████| 145/145 [15:03<00:00, 6.23s/it, loss=0.562]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 8/10 | Train Loss: 0.5618 | Val Loss: 0.5412 | Train Acc: 99.72% | Val Acc: 99.28%  
Seed 62 | Epoch 9/10: 100%|██████████| 145/145 [15:14<00:00, 6.31s/it, loss=0.555]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 9/10 | Train Loss: 0.5519 | Val Loss: 0.5323 | Train Acc: 99.82% | Val Acc: 99.28%  
Seed 62 | Epoch 10/10: 100%|██████████| 145/145 [15:06<00:00, 6.25s/it, loss=0.566]  
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(  
Seed 62 | Epoch 10/10 | Train Loss: 0.5429 | Val Loss: 0.5240 | Train Acc: 99.74% | Val Acc: 99.28%
```

```
Seed 62 completed in 154.38 minutes.
```

```
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted  
warnings.warn(
```

```
RESULTS FOR SEED 62
```

```
-----  
Test Accuracy   : 98.81%  
Test Sensitivity: 99.25%  
Test Specificity: 98.32%  
Test Precision  : 98.52%  
Test F1-Score   : 98.88%  
Test Error Rate : 1.19%  
Test ROC-AUC    : 0.9988
```

```
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```

```
STARTING FIVE-SEED EXPERIMENT – SEED 72
```

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=====
```

```
Random Seed: 72
```

```
=====
STARTING DVQC TRAINING FOR SEED 72
=====
Seed 72 | Epoch 1/10: 100%|██████████| 145/145 [14:44<00:00, 6.10s/it, loss=0.636]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 1/10 | Train Loss: 0.6420 | Val Loss: 0.6316 | Train Acc: 50.21% | Val Acc: 60.31%
Seed 72 | Epoch 2/10: 100%|██████████| 145/145 [15:13<00:00, 6.30s/it, loss=0.568]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 2/10 | Train Loss: 0.5978 | Val Loss: 0.6115 | Train Acc: 77.23% | Val Acc: 78.05%
Seed 72 | Epoch 3/10: 100%|██████████| 145/145 [15:13<00:00, 6.30s/it, loss=0.58]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 3/10 | Train Loss: 0.5816 | Val Loss: 0.5955 | Train Acc: 79.94% | Val Acc: 78.77%
Seed 72 | Epoch 4/10: 100%|██████████| 145/145 [15:12<00:00, 6.30s/it, loss=0.568]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 4/10 | Train Loss: 0.5656 | Val Loss: 0.5773 | Train Acc: 95.60% | Val Acc: 97.59%
Seed 72 | Epoch 5/10: 100%|██████████| 145/145 [15:16<00:00, 6.32s/it, loss=0.539]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 5/10 | Train Loss: 0.5468 | Val Loss: 0.5556 | Train Acc: 98.55% | Val Acc: 97.71%
Seed 72 | Epoch 6/10: 100%|██████████| 145/145 [15:17<00:00, 6.33s/it, loss=0.526]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 6/10 | Train Loss: 0.5236 | Val Loss: 0.5307 | Train Acc: 99.41% | Val Acc: 98.55%
Seed 72 | Epoch 7/10: 100%|██████████| 145/145 [15:19<00:00, 6.34s/it, loss=0.479]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 7/10 | Train Loss: 0.5026 | Val Loss: 0.5086 | Train Acc: 99.73% | Val Acc: 98.79%
Seed 72 | Epoch 8/10: 100%|██████████| 145/145 [15:17<00:00, 6.33s/it, loss=0.479]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 8/10 | Train Loss: 0.4852 | Val Loss: 0.4898 | Train Acc: 99.61% | Val Acc: 99.28%
Seed 72 | Epoch 9/10: 100%|██████████| 145/145 [15:07<00:00, 6.26s/it, loss=0.451]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 9/10 | Train Loss: 0.4670 | Val Loss: 0.4706 | Train Acc: 99.79% | Val Acc: 99.64%
Seed 72 | Epoch 10/10: 100%|██████████| 145/145 [15:04<00:00, 6.23s/it, loss=0.426]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 72 | Epoch 10/10 | Train Loss: 0.4502 | Val Loss: 0.4543 | Train Acc: 99.82% | Val Acc: 99.64%

Seed 72 completed in 154.70 minutes.
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

RESULTS FOR SEED 72

```
-----
Test Accuracy : 99.01%
Test Sensitivity: 99.63%
Test Specificity: 98.32%
Test Precision : 98.52%
Test F1-Score : 99.07%
Test Error Rate : 0.99%
Test ROC-AUC : 0.9964
```

===== STARTING FIVE-SEED EXPERIMENT – SEED 82 =====

Random Seed: 82

```
=====
STARTING DVQC TRAINING FOR SEED 82
=====
Seed 82 | Epoch 1/10: 100%|██████████| 145/145 [14:59<00:00, 6.21s/it, loss=0.609]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 1/10 | Train Loss: 0.6431 | Val Loss: 0.6334 | Train Acc: 83.55% | Val Acc: 95.54%
Seed 82 | Epoch 2/10: 100%|██████████| 145/145 [14:52<00:00, 6.16s/it, loss=0.612]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 2/10 | Train Loss: 0.6115 | Val Loss: 0.6161 | Train Acc: 99.22% | Val Acc: 97.95%
Seed 82 | Epoch 3/10: 100%|██████████| 145/145 [14:51<00:00, 6.15s/it, loss=0.592]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 3/10 | Train Loss: 0.5967 | Val Loss: 0.6022 | Train Acc: 99.65% | Val Acc: 98.67%
Seed 82 | Epoch 4/10: 100%|██████████| 145/145 [14:47<00:00, 6.12s/it, loss=0.578]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
```

```
07/09/2026, 18:39          EFFICIENTNET-B0 + 8 DVQC # FIVE-SEED ABLATION EXPERIMENT.ipynb - Colab
Seed 82 | Epoch 4/10 | Train Loss: 0.5825 | Val Loss: 0.5873 | Train Acc: 99.76% | Val Acc: 98.79%
Seed 82 | Epoch 5/10: 100%[████████] 145/145 [14:49<00:00, 6.14s/it, loss=0.553]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 5/10 | Train Loss: 0.5679 | Val Loss: 0.5713 | Train Acc: 99.82% | Val Acc: 99.64%
Seed 82 | Epoch 6/10: 100%[████████] 145/145 [14:49<00:00, 6.14s/it, loss=0.562]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 6/10 | Train Loss: 0.5525 | Val Loss: 0.5565 | Train Acc: 99.85% | Val Acc: 99.03%
Seed 82 | Epoch 7/10: 100%[████████] 145/145 [14:54<00:00, 6.17s/it, loss=0.524]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 7/10 | Train Loss: 0.5369 | Val Loss: 0.5389 | Train Acc: 99.82% | Val Acc: 99.76%
Seed 82 | Epoch 8/10: 100%[████████] 145/145 [15:01<00:00, 6.22s/it, loss=0.514]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 8/10 | Train Loss: 0.5210 | Val Loss: 0.5242 | Train Acc: 99.83% | Val Acc: 99.52%
Seed 82 | Epoch 9/10: 100%[████████] 145/145 [14:49<00:00, 6.14s/it, loss=0.48]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 9/10 | Train Loss: 0.5045 | Val Loss: 0.5092 | Train Acc: 99.92% | Val Acc: 99.40%
Seed 82 | Epoch 10/10: 100%[████████] 145/145 [15:01<00:00, 6.22s/it, loss=0.475]
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(
Seed 82 | Epoch 10/10 | Train Loss: 0.4886 | Val Loss: 0.4959 | Train Acc: 99.96% | Val Acc: 98.91%
```

Seed 82 completed in 151.70 minutes.
/usr/local/lib/python3.13/dist-packages/PIL/Image.py:1047: UserWarning: Palette images with Transparency expressed in bytes should be converted
warnings.warn(

RESULTS FOR SEED 82

Test Accuracy : 98.22%
Test Sensitivity: 98.13%
Test Specificity: 98.32%
Test Precision : 98.50%
Test F1-Score : 98.32%
Test Error Rate : 1.78%
Test ROC-AUC : 0.9933

INDIVIDUAL FIVE-SEED RESULTS

	Seed	Split	Accuracy (%)	Sensitivity (%)	Specificity (%)	Precision (%)	F1-Score (%)	Error Rate (%)	ROC-AUC
0	42	Training	99.870214	99.978448	99.761181	99.763441	99.870829	0.129786	0.999956
1	42	Validation	99.879373	99.796748	100.000000	100.000000	99.898271	0.120627	0.999023
2	42	Testing	98.814229	99.626866	97.899160	98.161765	98.888889	1.185771	0.998667
3	52	Training	99.902661	99.870690	99.934868	99.935303	99.902986	0.097339	0.999964
4	52	Validation	99.155609	98.983740	99.406528	99.591002	99.286442	0.844391	0.999445
5	52	Testing	98.221344	98.134328	98.319328	98.501873	98.317757	1.778656	0.996676
6	62	Training	99.924292	99.956897	99.891446	99.892311	99.924593	0.075708	0.999949
7	62	Validation	99.276236	99.593496	98.813056	99.190283	99.391481	0.723764	0.999349
8	62	Testing	98.814229	99.253731	98.319328	98.518519	98.884758	1.185771	0.998777
9	72	Training	99.967554	99.978448	99.956578	99.956906	99.967676	0.032446	0.999999
10	72	Validation	99.638118	99.796748	99.406528	99.594320	99.695431	0.361882	0.999831
11	72	Testing	99.011858	99.626866	98.319328	98.523985	99.072356	0.988142	0.996363
12	82	Training	99.935107	99.892241	99.978289	99.978430	99.935317	0.064893	0.999942
13	82	Validation	98.914355	98.983740	98.813056	99.185336	99.084435	1.085645	0.998842
14	82	Testing	98.221344	98.134328	98.319328	98.501873	98.317757	1.778656	0.993337

DVQC FIVE-SEED MEAN ± STANDARD DEVIATION

	Split	Metric	Mean	Standard Deviation	Mean ± SD
0	Training	Accuracy (%)	99.919965	0.036357	99.92 ± 0.04
1	Training	Sensitivity (%)	99.935345	0.050543	99.94 ± 0.05

	Split	Metric	Mean	Standard Deviation	Mean ± SD
2	Training	Specificity (%)	99.904472	0.086299	99.90 ± 0.09
3	Training	Precision (%)	99.905278	0.085446	99.91 ± 0.09
4	Training	F1-Score (%)	99.920280	0.036179	99.92 ± 0.04
5	Training	Error Rate (%)	0.080035	0.036357	0.08 ± 0.04
6	Training	ROC-AUC	0.999962	0.000022	1.00 ± 0.00
7	Validation	Accuracy (%)	99.372738	0.385253	99.37 ± 0.39
8	Validation	Sensitivity (%)	99.430894	0.416543	99.43 ± 0.42
9	Validation	Specificity (%)	99.287834	0.496534	99.29 ± 0.50
10	Validation	Precision (%)	99.512188	0.339622	99.51 ± 0.34
11	Validation	F1-Score (%)	99.471212	0.325095	99.47 ± 0.33
12	Validation	Error Rate (%)	0.627262	0.385253	0.63 ± 0.39
13	Validation	ROC-AUC	0.999298	0.000385	1.00 ± 0.00
14	Testing	Accuracy (%)	98.616601	0.369729	98.62 ± 0.37
15	Testing	Sensitivity (%)	98.955224	0.764698	98.96 ± 0.76
16	Testing	Specificity (%)	98.235294	0.187905	98.24 ± 0.19
17	Testing	Precision (%)	98.441603	0.156746	98.44 ± 0.16
18	Testing	F1-Score (%)	98.696303	0.353771	98.70 ± 0.35
19	Testing	Error Rate (%)	1.383399	0.369729	1.38 ± 0.37
20	Testing	ROC-AUC	0.996764	0.002213	1.00 ± 0.00

=====

FINAL TEST-SET MEAN ± STANDARD DEVIATION

=====

	Metric	Mean	Standard Deviation	Mean ± SD
14	Accuracy (%)	98.616601	0.369729	98.62 ± 0.37
15	Sensitivity (%)	98.955224	0.764698	98.96 ± 0.76
16	Specificity (%)	98.235294	0.187905	98.24 ± 0.19
17	Precision (%)	98.441603	0.156746	98.44 ± 0.16
18	F1-Score (%)	98.696303	0.353771	98.70 ± 0.35
19	Error Rate (%)	1.383399	0.369729	1.38 ± 0.37
20	ROC-AUC	0.996764	0.002213	1.00 ± 0.00

=====

TEST CONFUSION MATRICES FOR FIVE SEEDS

=====

Seed 42

```
[[233  5]
 [ 1 267]]
```

Seed 52

```
[[234  4]
 [ 5 263]]
```

Seed 62

```
[[234  4]
 [ 2 266]]
```

Seed 72

```
[[234  4]
 [ 1 267]]
```

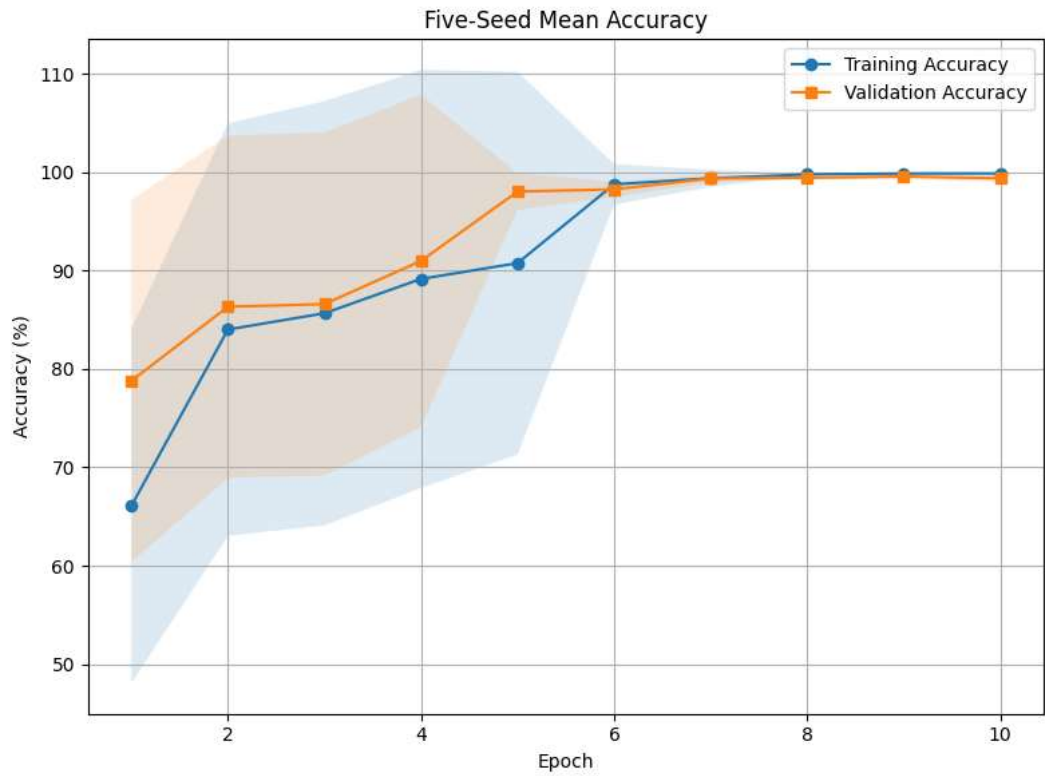
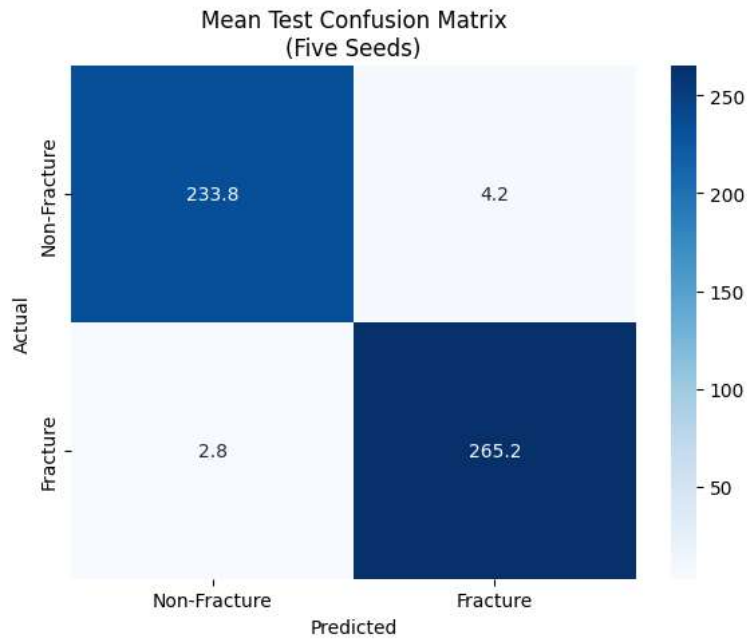
Seed 82

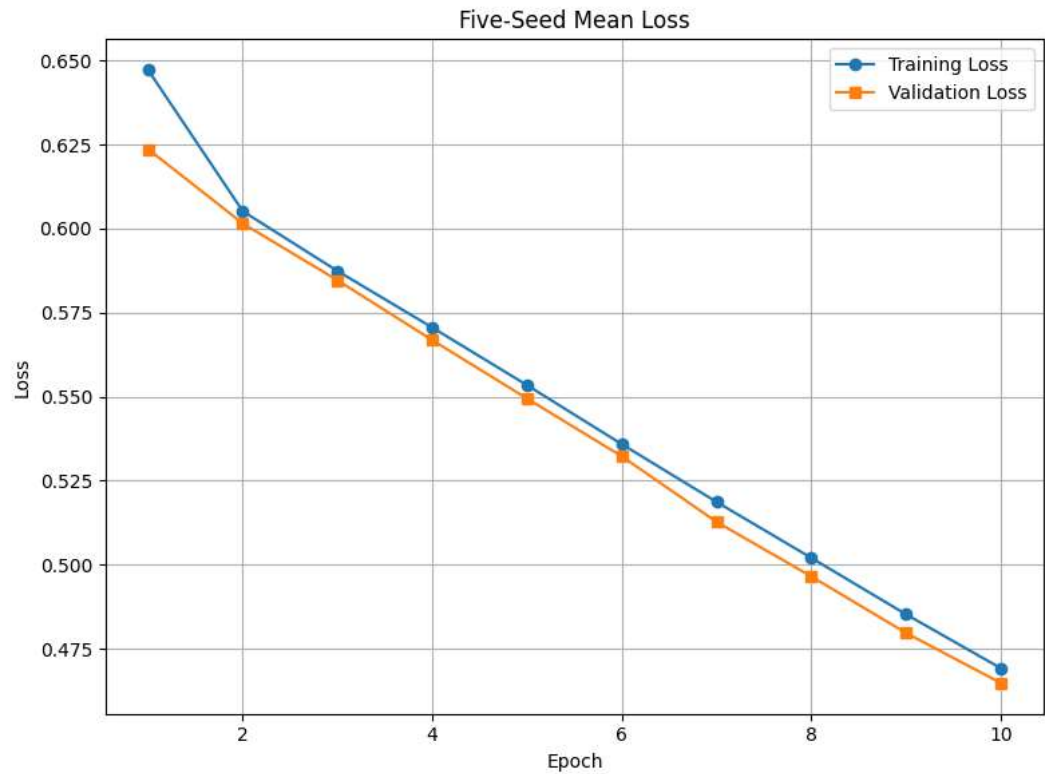
```
[[234  4]
 [ 5 263]]
```

=====

MEAN TEST CONFUSION MATRIX

```
[[233.8  4.2]
 [ 2.8 265.2]]
```





```
=====
FINAL VERIFICATION
=====
✓ Five independent seeds used:
  [42, 52, 62, 72, 82]
✓ Training metrics calculated
✓ Validation metrics calculated
✓ Testing metrics calculated
✓ Accuracy calculated
✓ Sensitivity calculated
✓ Specificity calculated
✓ Precision calculated
✓ F1-Score calculated
✓ Error Rate calculated
✓ ROC-AUC calculated
✓ Mean calculated
✓ Standard Deviation calculated
✓ Mean ± SD calculated
✓ Individual seed results saved
✓ Mean ± SD results saved

Saved files:
1. DVQC_8Qubit_5Seed_Individual_Results.csv
2. DVQC_8Qubit_Mean_SD_Results.csv

Experiment completed successfully.
```